Unnamed_Unit

Unit 2 - Bringing Ideas to Life

PDF Documents

PDF Document 1: 1738151089823-Chapter 9 - Unit 2 - Bringing Ideas to Life.pdf

This PDF document is included in the unit materials.

UNIT 2 BRINGING IDEAS TO LIFE

Project Introduction

INTRODUCTION

In this unit, you will design an Eco-Friendly Traffic Light System that not only controls traffic efficiently but also incorporates renewable energy sources. Building on the Button-Controlled Traffic Light concept, this project adds an environmental module to simulate a solar panel, providing a sustainable power source. Additionally, the system will include a display module to show the energy level available to run the traffic light, which changes based on the amount of light detected by the "solar panel."

IDEA GENERATION

Start by brainstorming how your Eco-Friendly Traffic Light System should function, considering the following aspects:

ENERGY GENERATION AND MANAGEMENT

Think about how the system will simulate energy collection through the environment module, acting as a solar panel. The system should be able to detect light and increase the energy level displayed on the screen.

TRAFFIC CONTROL MECHANISM

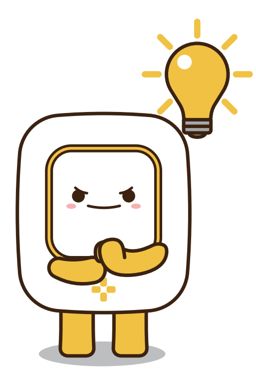
Like in the Button-Controlled Traffic Light, the traffic lights will change based on user input. But in this variation, the system should also consider the available energy when determining how the lights operate.

ENERGY DISPLAY

The display module will show the current energy level, letting users know how much power is available to keep the traffic light system running. If enough light is detected for a certain period, the energy level goes up.

TIPS

Think about how real-life solar panels work and how energy storage is displayed in devices. Your system should balance energy usage with energy generation to ensure the traffic lights can operate sustainably.

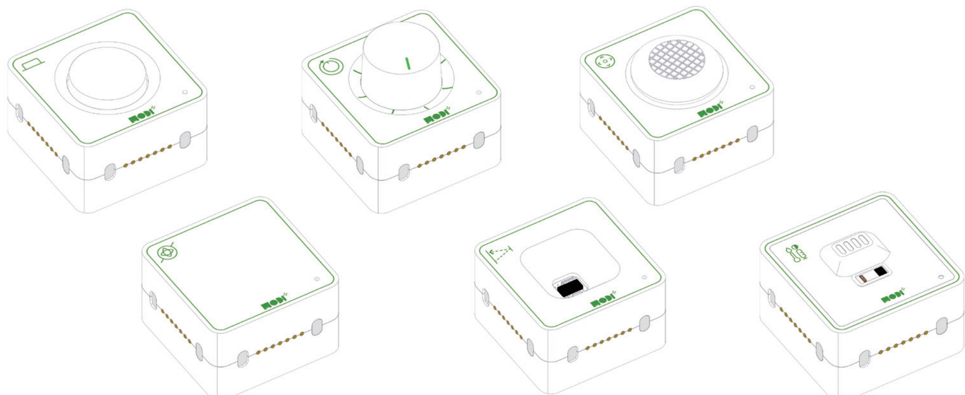


Required Modules & Brainstorming

To build your Eco-Friendly Traffic Light System, you'll need to carefully select which MODI modules will best fit your project. Here are some hints to guide your brainstorming process:

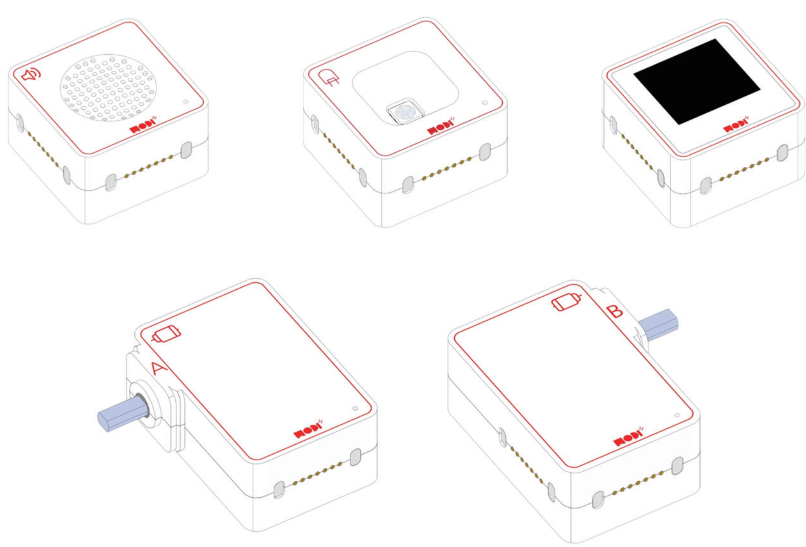
INPUT MODULES

- **Detecting Environmental Conditions:** Consider a sensor that can measure external environmental factors, such as light levels, to simulate how much energy is being generated for your system. This module will help determine how the system reacts to changing conditions throughout the day.



OUTPUT MODULES

- **Displaying Information:** The system needs to show the energy level and traffic light status clearly. What module could present this information in a way that's easy to understand and helps users know how the traffic system is operating?
- **Traffic Light Control:** Reflect on how the system will visually signal the status of the traffic lights to drivers and pedestrians. What module could be used to show red, yellow, or green lights effectively?



Required Modules & Brainstorming

Which MODI modules do you need for the creation?

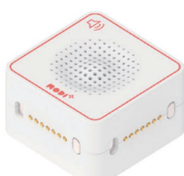
DIAL

BUTTON

ENVIRONMENT

JOYSTICK

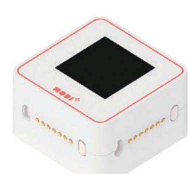
TOF



SPEAKER

IMU

LED



DISPLAY

MOTOR A

MOTOR B

HOW DOES THE CREATION LOOK LIKE?